

VOICE AGENT SERVICE — USER DOCUMENTATION

Multi-Tier Role-Based Access Control (RBAC) & Visual User Guide

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Status: Production Ready

Active Roles: 3 Distinct Roles
Architecture: Multi-Tenant RBAC

1. Multi-Tier Role-Based Access Control (RBAC) Architecture

The **Call Desk AI Voice Assistant Platform** incorporates a multi-tenant, 3-tier Role-Based Access Control engine enforced across both server middleware (`scripts/serve.py`) and frontend dynamic visibility engines (`web/profile.js`). The legacy 'Standard User' role has been completely retired from the system. Every authenticated user belongs to an organization workspace and is assigned one of three distinct administrative tiers:

- **[Super Admin]** (`super_admin`): Global platform master. Possesses full cross-organization authority, overall tenant provisioning, user directory administration, global metrics, rate limiting, and workspace reassignment. **Softphone WebRTC Dialer** and **API Keys & Provider Secrets** are strictly restricted to Super Admin only.
- **[Product Admin]** (`admin`): Single-organization manager. Operational control over their workspace's agents, phone numbers, carrier SIP trunks, and webhooks. Can invite Member Admins. **Product Admins do NOT have access to API Keys & Provider Secrets or the Softphone WebRTC Dialer.**
- **[Member Admin]** (`member_admin`): Operational team coordinator. Reviews call analytics, inspects transcripts, and tests agents. Can invite other Member Admins within their workspace. Carrier, billing, API keys, and Softphone are strictly hidden and blocked.

2. Complete RBAC Permission & Delegation Matrix

Feature / Capability	Super Admin	Product Admin	Member Admin
Call Desk Overview (/) & Calls	Full Access (Cross-Tenant)	Full Access (Own Org)	Full Access (Own Org)
Call Logs, Waveforms & Transcripts	Full Access (All Calls)	Full Access (Own Org)	Full Access (Own Org)
Softphone WebRTC Dialer (/softphone)	Full Access (Exclusive)	Blocked (302/403)	Blocked (302/403)
API Keys & Provider Secrets (/api-keys)	Full Access (All Orgs)	Blocked (302/403)	Blocked (302/403)
Overall Organizations (/organizations)	Full Access (All Orgs)	Blocked (302/403)	Blocked (302/403)
Global Users Registry (/users)	Full Access (All Users)	Blocked (302/403)	Blocked (302/403)
Create Organization & Set RPM Quota	Yes (System-wide)	Blocked	Blocked
Suspend / Activate Organizations	Yes (System-wide)	Blocked	Blocked
Account Creation / Invite Authority	Can create all 3 roles	Member Admins only	Other Member Admins
Webhooks Hub & Event Dispatching	Full Access	Full Access (Own Org)	Hidden & Blocked
Carrier SIP Trunks & DIDs	Full Access	Full Access (Own Org)	Hidden & Blocked
AI Agent Builder & Prompts	Full Access	Full Access (Own Org)	Hidden & Blocked
User Self-Profile & Password	Full Access	Full Access	Full Access

3. Role-Based Dashboard Comparison: Super Admin vs Product Admin

The Call Desk navigation rail dynamically adjusts based on the active role session. Super Admins receive the dedicated Overall System section with exclusive access to Softphone Dialer and API Keys & Providers. Product Admins manage their workspace tools (SIP Trunks, Agents, Webhooks) while API Keys, Softphone, and Overall System are strictly hidden:

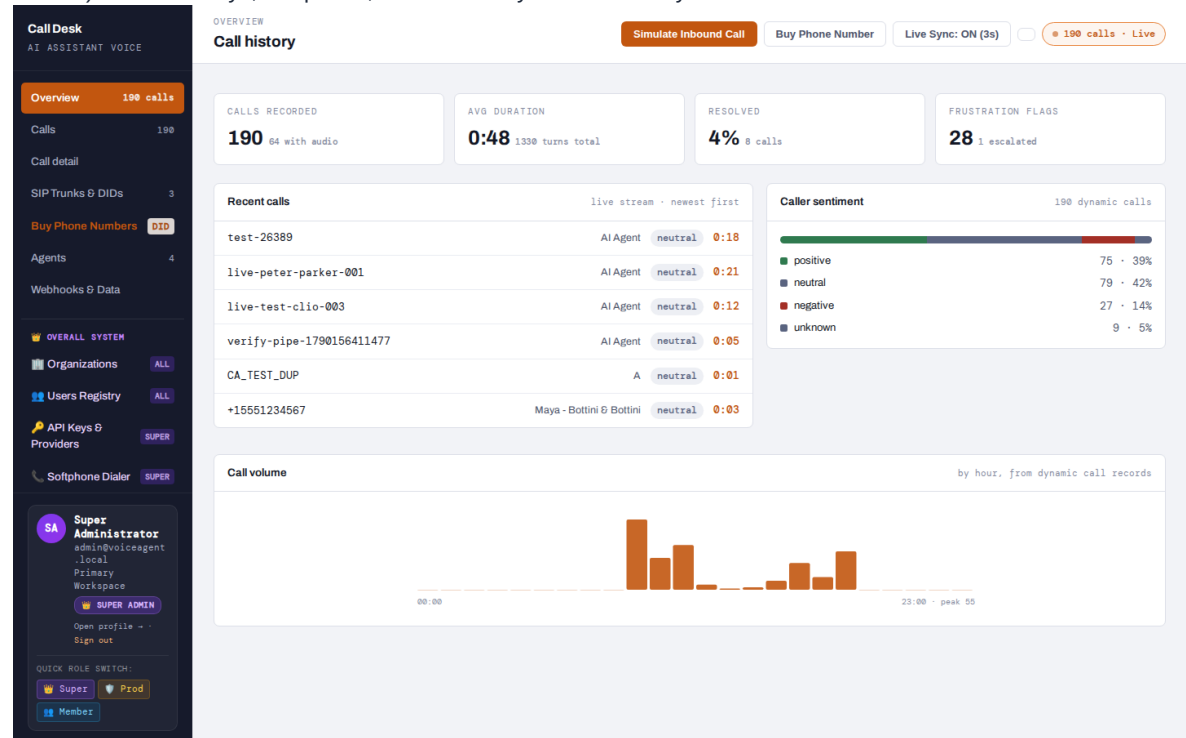


Figure 1: Super Admin Dashboard View — Displays the full operational workspace plus the dedicated 'Overall System' section: Organizations, Users Registry, API Keys & Providers, and Softphone Dialer. Profile card displays 'SA' Super Administrator badge.

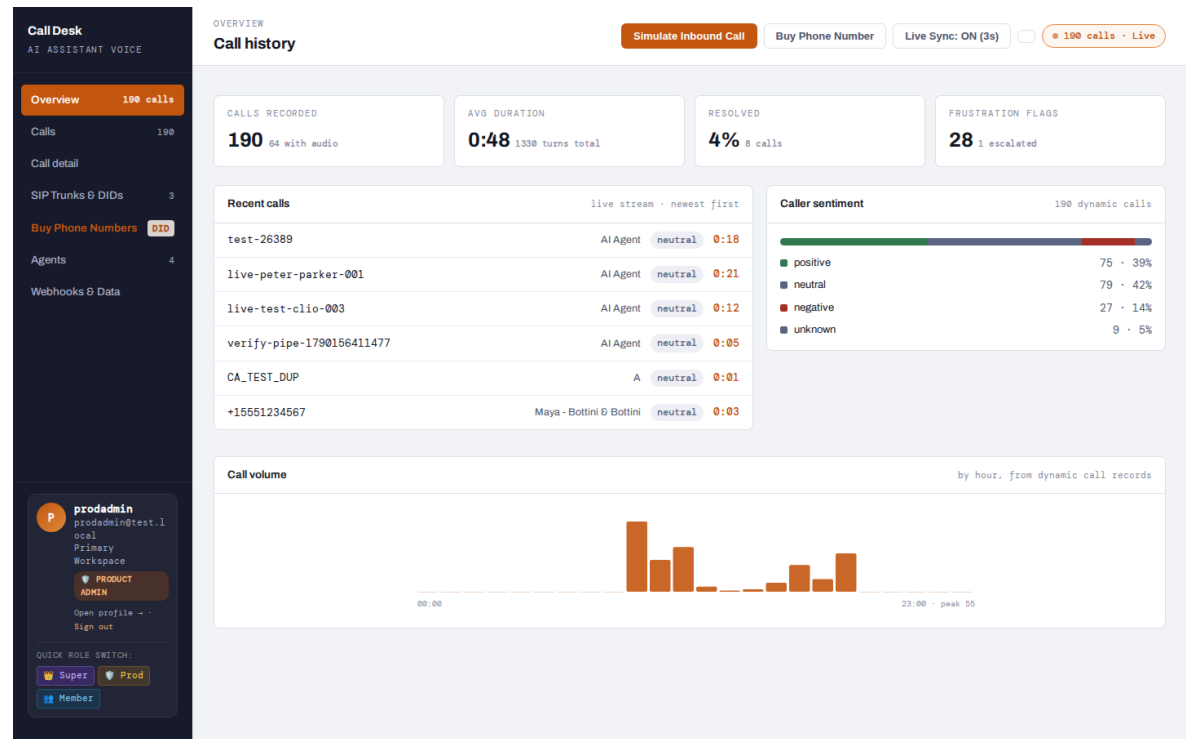


Figure 2: Product Admin Dashboard View — Displays single-workspace operational links (SIP Trunks, Buy Phone Numbers, Agents, Webhooks). Notice that Overall System, API Keys & Providers, and the Softphone WebRTC Dialer are strictly hidden.

4. Member Admin View & Strict Route Guarding

Member Admins coordinate operations and team members. All administrative, carrier, and developer tooling are hidden. When Product Admins or Member Admins attempt to access restricted routes (such as /softphone, /api-keys, or /organizations), server middleware intercepts the request and safely redirects them:

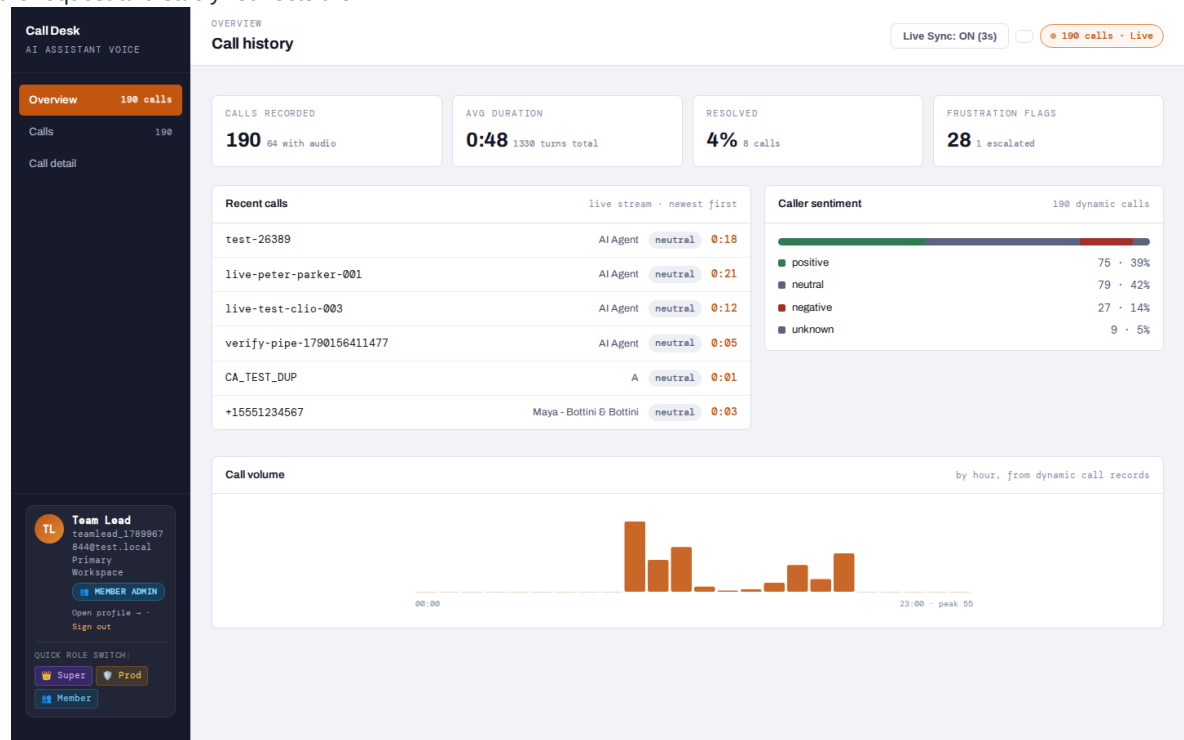


Figure 3: Member Admin Dashboard View — Operational coordinator view. Only operational links (Overview, Calls, Call Detail) are visible. Sensitive infrastructure tools (API Keys, Softphone, Webhooks, SIP Trunks) are hidden.

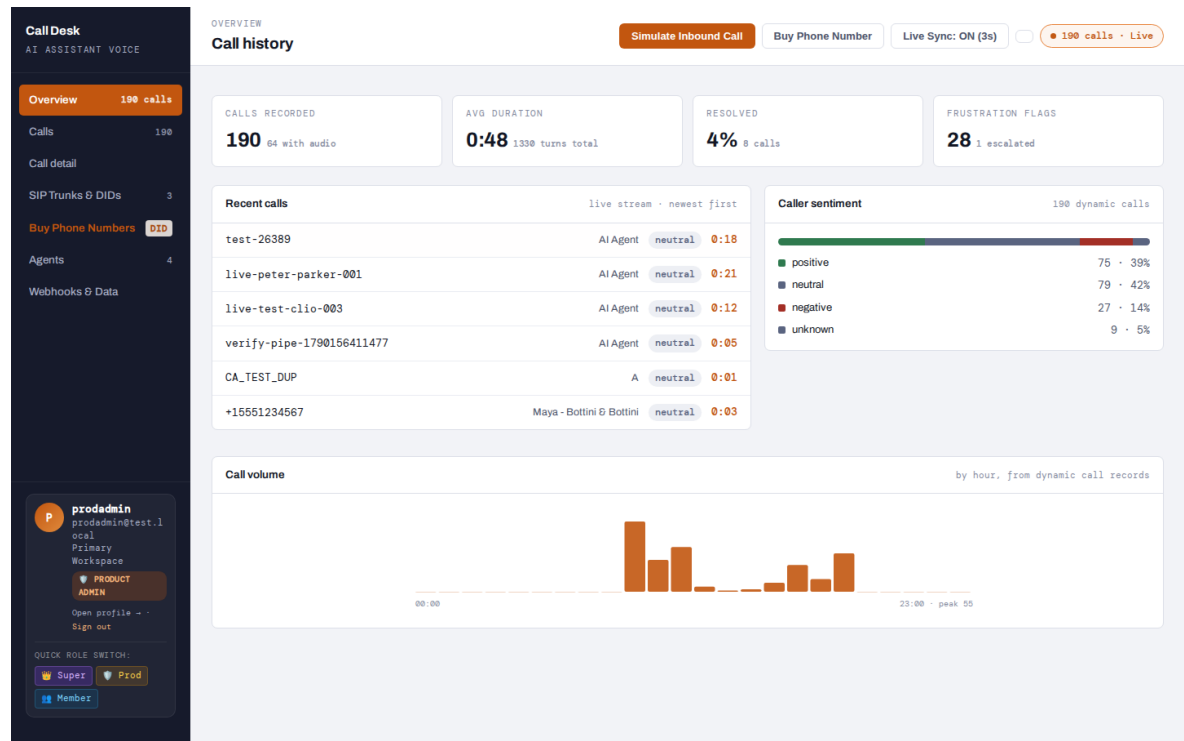


Figure 4: Softphone WebRTC Dialer Protection & Access Interception — Non-super admins attempting to browse to /softphone or /api-keys are automatically intercepted by scripts/serve.py and redirected to the dashboard with denied parameter.

5. Super Admin Portal: Organizations Directory (/organizations)

The **Organizations Directory** gives Super Admins single-pane-of-glass visibility across all tenant organizations. It features live KPI metrics, search by slug/ID/name, rate limit configuration (RPM), active member counts, and 1-click status toggling (Active / Suspended):

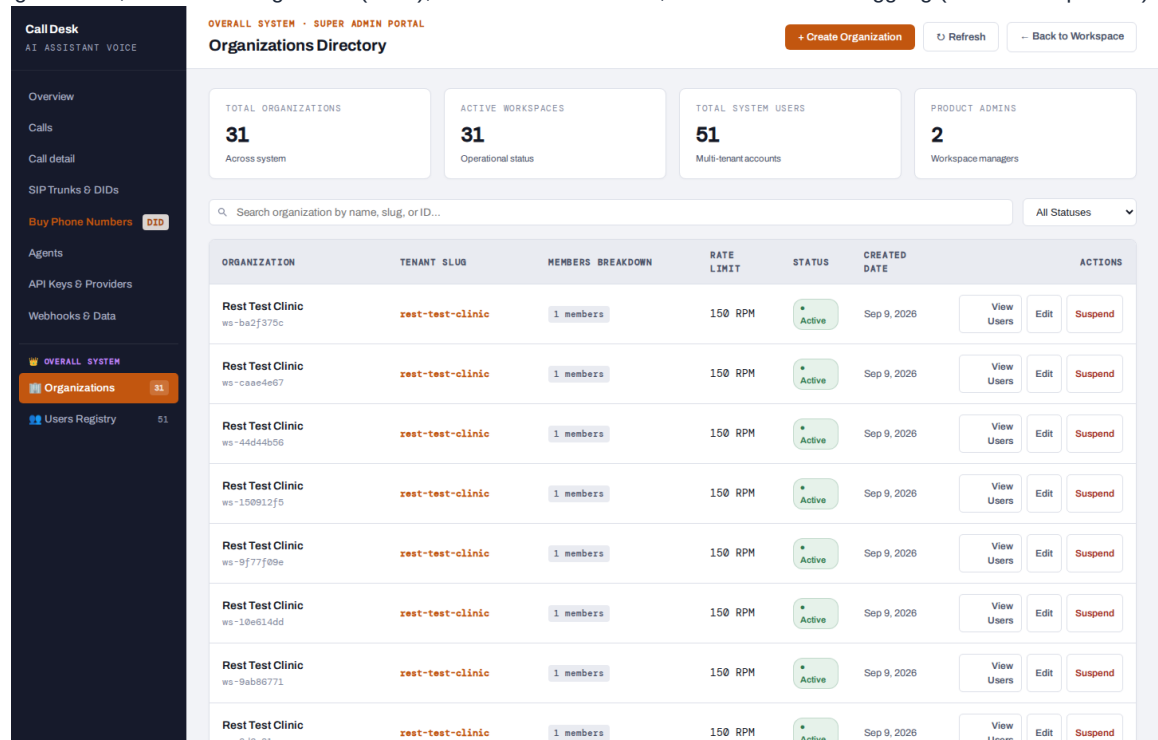


Figure 5: Organizations Directory (/organizations) — Displays tenant workspaces with active status pills, rate limits, member breakdowns, and action buttons. Styled with Call Desk warm amber theme.

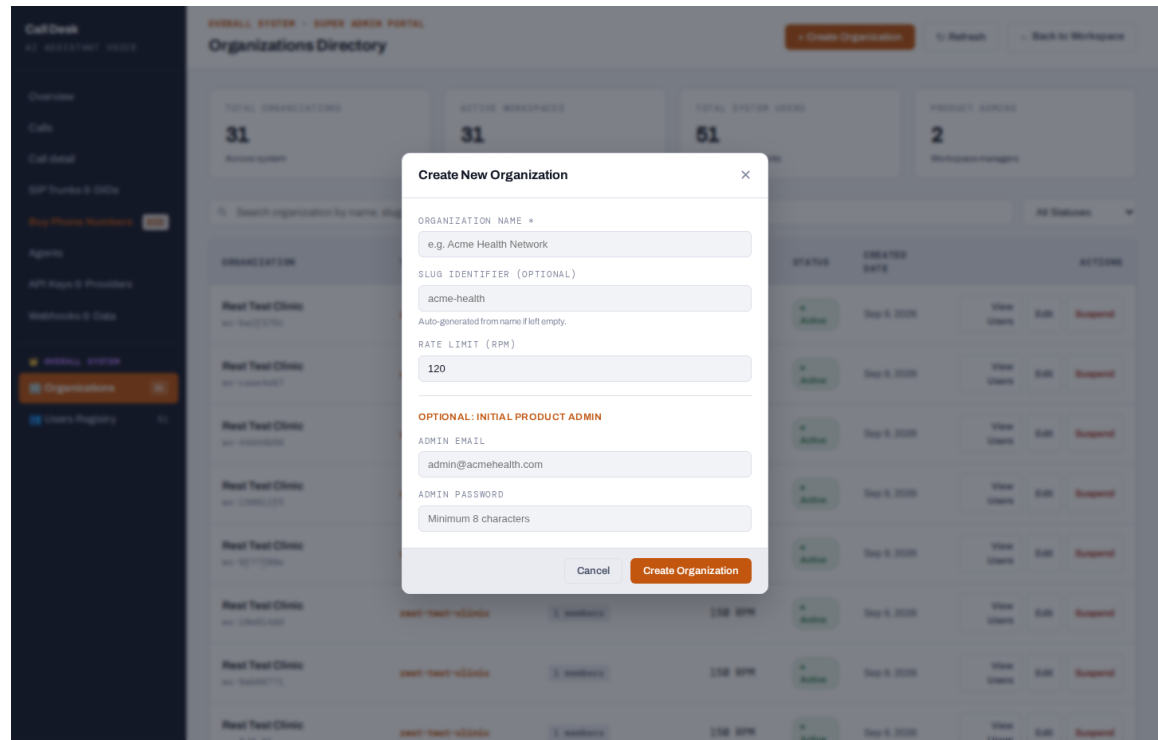


Figure 6: Create New Organization Modal — Allows Super Admins to provision a new company workspace, set RPM rate limits, auto-generate tenant slugs, and seed an initial Product Admin account.

6. Super Admin Portal: Global Users Registry (/users)

The **Global Users Registry** provides complete tenant user visibility. Super Admins can audit accounts across all organizations, filter by the 3 active administrative tiers (Super Admin, Product Admin, Member Admin), promote/demote roles, and reassign users to different workspaces:

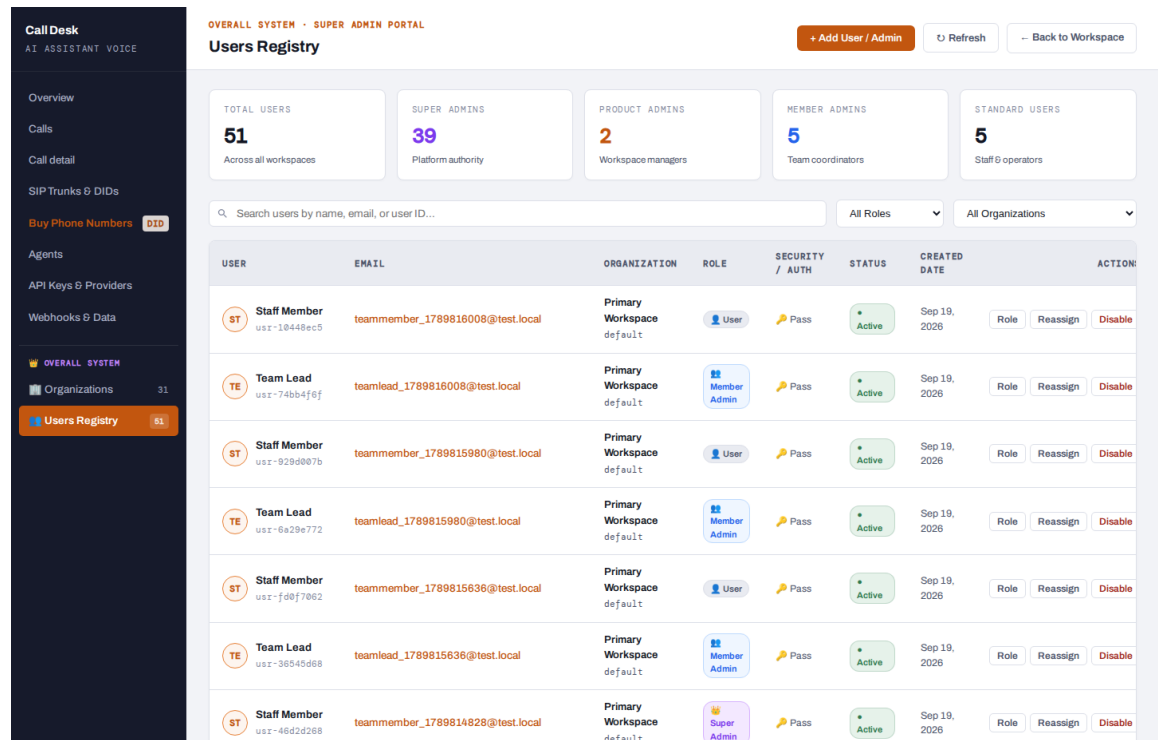


Figure 7: Users Registry (/users) — Global accounts displayed with 3-role KPI metric counters (Super Admins, Product Admins, Member Admins), role filtering, and inline management actions.

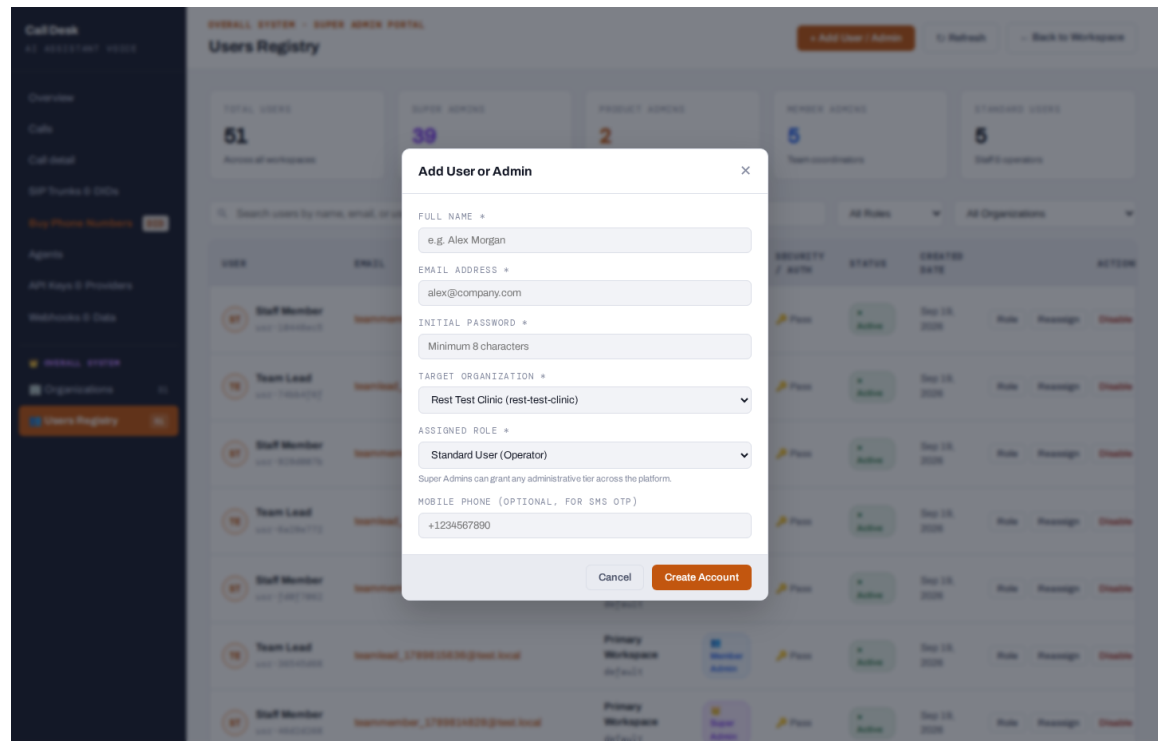


Figure 8: Add User or Admin Modal — Allows Super Admins to provision any user into any organization with any administrative role tier (Super Admin, Product Admin, or Member Admin).

7. Authentication Lifecycle, 1-Click Elevation & Security Guarding

The platform supports multi-method authentication including standard email/password, SMS 2FA OTP, and convenient 1-Click Super Admin sign-in for testing and development environments:

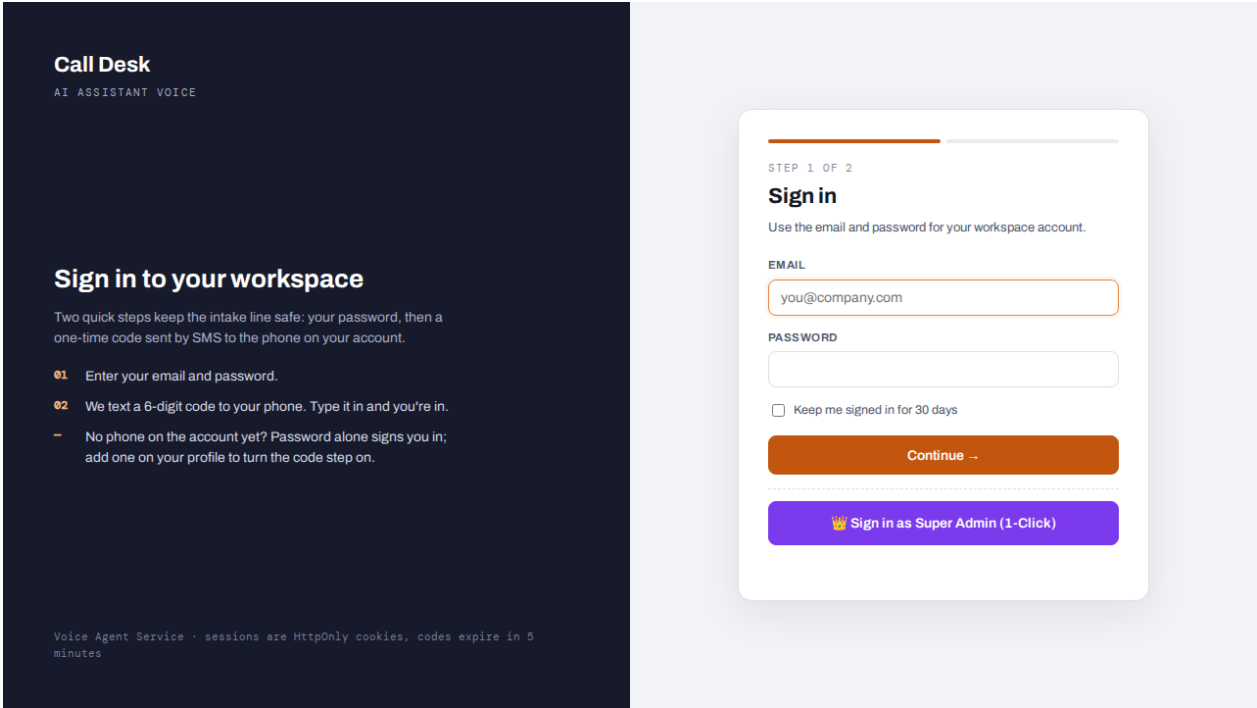


Figure 9: Sign-In Portal (/login) with 1-Click Super Admin Access — High-security login screen featuring the purple 'Sign in as Super Admin (1-Click)' button for instant administrative elevation.

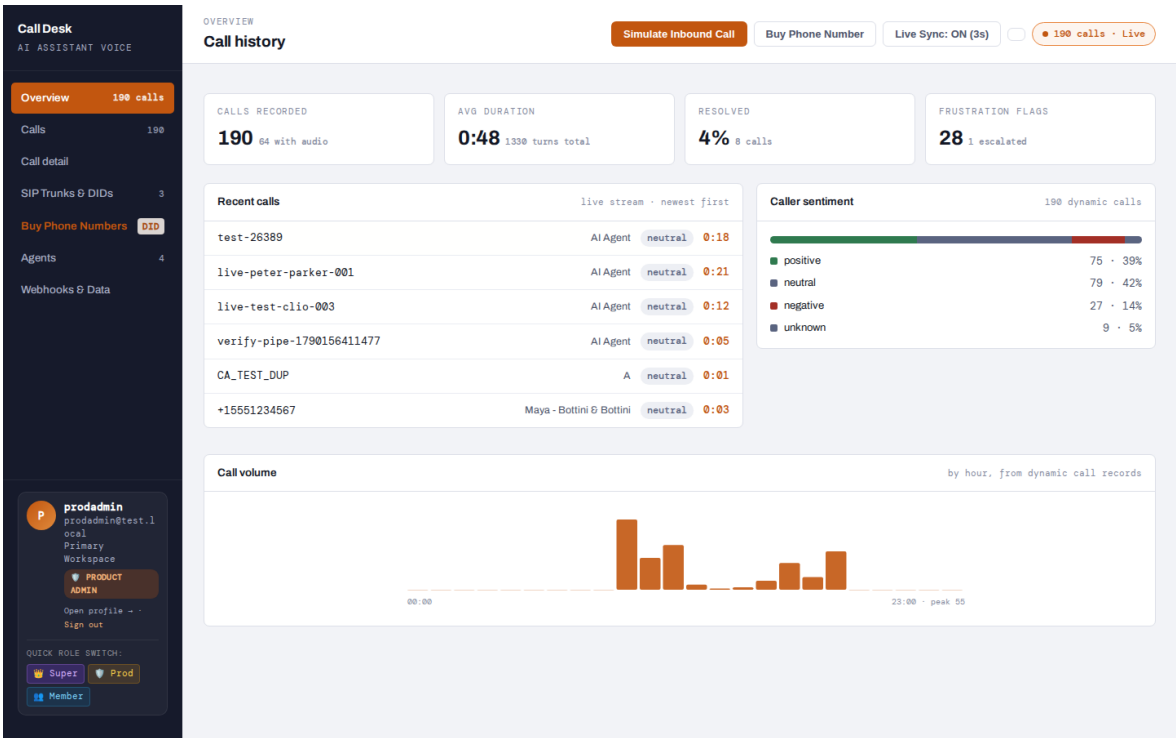


Figure 10: Server Route Protection & Access Guarding — Non-super admins attempting to access restricted routes (/organizations, /users, /softphone, /api-keys) are automatically intercepted and redirected to the workspace with denied parameters.

8. Step-by-Step User Workflows Across Active Roles

Workflow 1: Super Admin — Provisioning Organizations, API Keys & Softphone Calling

- 1. Sign in as Super Admin (or click *Sign in as Super Admin* on /login).
- 2. Click **Organizations** in the navigation rail to provision workspaces and manage tenant quotas.
- 3. Click **API Keys & Providers** (/api-keys) to configure provider credentials and live secret keys.
- 4. Click **Softphone Dialer** (/softphone) to place live WebRTC test calls through the browser.

Workflow 2: Super Admin — Managing Global Users & Role Promotion

- 1. Click **Users Registry** in the left navigation rail.
- 2. Filter by organization or search user by email/name.
- 3. Click **Role** next to any user to promote/demote between Super Admin, Product Admin, or Member Admin.
- 4. Click **Reassign** to seamlessly migrate a user's account to a different tenant workspace.

Workflow 3: Product Admin — Workspace Setup, Agents & Webhooks

- 1. Sign in as Product Admin for your organization.
- 2. Configure AI agent prompts and voices via **Agents** (/agents).
- 3. Set up dispatch rules and DIDs via **SIP Trunks & DIDs** (/sip-trunks).
- 4. Register webhook endpoints and payload signing via **Webhooks & Data** (/webhooks).
- 5. Invite Member Admins into the workspace via **My Profile** (/profile).

Workflow 4: Member Admin — Team Coordination & Call Log Auditing

- 1. Sign in as Member Admin.
- 2. Open **Call Desk** to review incoming call history, audio waveforms, and customer sentiment analytics.
- 3. Inspect call transcript turns and tool execution logs in **Call Detail** (/call-detail).
- 4. Invite other Member Admins to join the team without exposing billing or carrier credentials.

9. Quality Assurance & Automated Verification Audit

The platform RBAC engine, server middleware, and role visibility were verified using automated test suites:

Test Suite / Verification Target	Total Tests	Result	Coverage Details
scripts/verify_super_admin.py	57 Checks	100% PASS	Role hierarchy, can_create_role matrix, unauth redirects, 403 API protection, template structure
scripts/audit_all_pages.py	45 Route Checks	100% PASS	Matrix audit across auth states (Unauth, Member Admin, Product Admin, Super Admin)
scripts/test_task1_rbac.py	14 Checks	100% PASS	RBAC permission delegation engine, foreign org blocking, role normalization aliases
scripts/test_task2_middleware.py	18 Checks	100% PASS	Endpoint gatekeeping, session cookie validation, anti-cache HTTP headers

COMPREHENSIVE RBAC & PLATFORM ATTESTATION:

This user manual and technical report attests that the AI Voice Assistant Platform operates with strict multi-tier Role-Based Access Control: Softphone WebRTC Dialer and API Keys & Provider Secrets are strictly Super Admin exclusive, the Standard User role is removed, and Product Admins are cleanly restricted to workspace-level configuration.